

RavenStack – Updating Reactive Customer Success Operations

At a glance

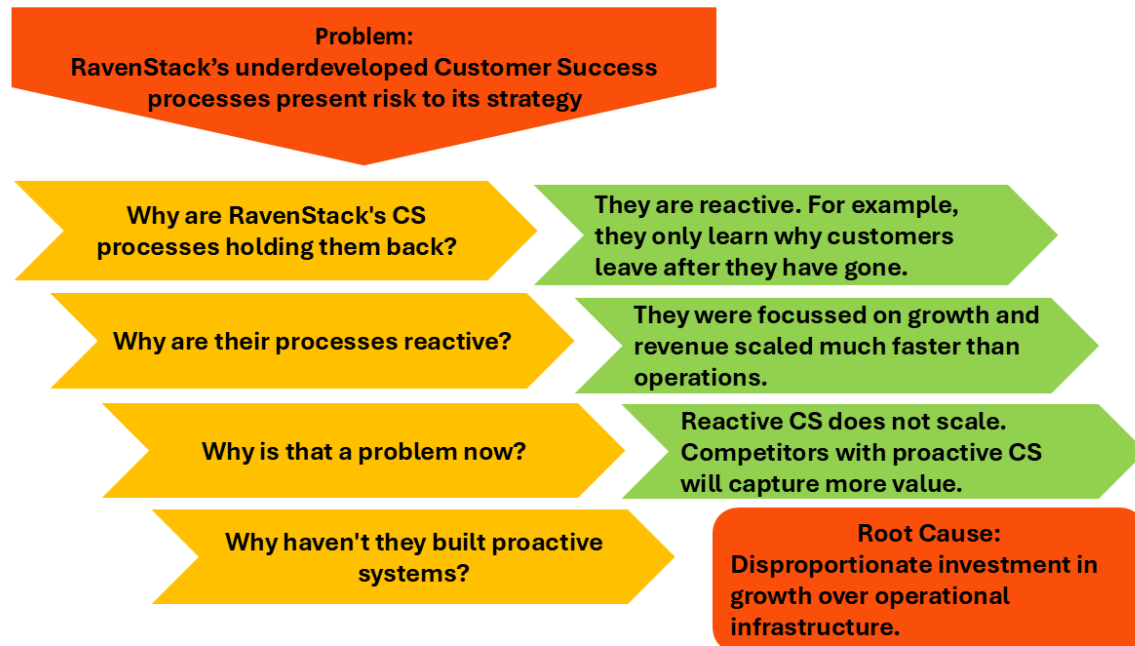
- **Segmentation** — Two customer clusters identified: Premier (78 accounts) and Core (422). Built from revenue and license count data; firmographic data was not detailed enough to use.
- **Health Scoring** — Shipped a v1 scoring framework combining product usage and support quality. Scores cluster in a narrow range due to source data limitations. Core accounts with unusually low scores flagged for review.
- **Churn Prediction** — Shipped a v1 churn model, predicting correctly 57% of the time against a 77% baseline. Weak signals in the data limit performance. Despite this, operational recommendations target over \$7M in at-risk revenue. Recommendations for continuous data improvements are documented.

Introduction

RavenStack is a SaaS business that has grown rapidly to \$122M ARR. Early on they implemented reactive Customer Success (CS) systems, such as capturing churn events. Two years of accelerated growth resulted in systems and processes that leadership believe are holding them back now.

They researched alignment of CS processes with revenue and their business plan. Using the white paper 'The SaaS Midlife Crisis: Navigating Customer Success Transformation as Your Company Scales Beyond Initial Hypergrowth' (Weiss, 2025) they verified their maturity as Series D measured in revenue. But it revealed systems and process gaps that positions them as a Series B organisation. Closing these gaps is now their priority.

To start this process, data has been shared with an analyst to carry out exploratory, explanatory and predictive analysis with the goal of identifying what can be implemented now and what requires further investment.



Five Whys analysis: diagnosing the root cause of RavenStack's operational gap.

Problem

RavenStack's rapid growth has outpaced its operational infrastructure. The result is a reactive, undifferentiated operation with no predictive or prescriptive abilities.

Objectives

1. Explore indicators within the data to develop a customer segmentation framework
2. Build a health scoring framework that gives CSMs a single view of account status
3. Explore churn data and attempt to predict customer churn
4. Deliver operational recommendations to optimise the value of existing customer relationships
5. Where limitations exist, document and/or workaround to provide RavenStack a roadmap aligned with best practices

Data and approach

The analysis used five internal data sources:

- **Accounts:** 500 records, identifiers, revenue, churn flags, rudimentary firmographics
- **Subscriptions:** 5000 records, granular breakdown of revenue, plan/tier
- **Support Tickets:** 2000 tickets, response/resolution metrics, satisfaction scores
- **Usage Metrics:** 25000 records, feature count/duration metrics
- **Churn Events:** 600 records, pre-churn activity, reason codes, customer feedback

I applied k-means clustering to ARR and seat counts, engineered a composite health score from Usage and Support data and trained a decision tree classifier to predict churn.

Results

Segmentation

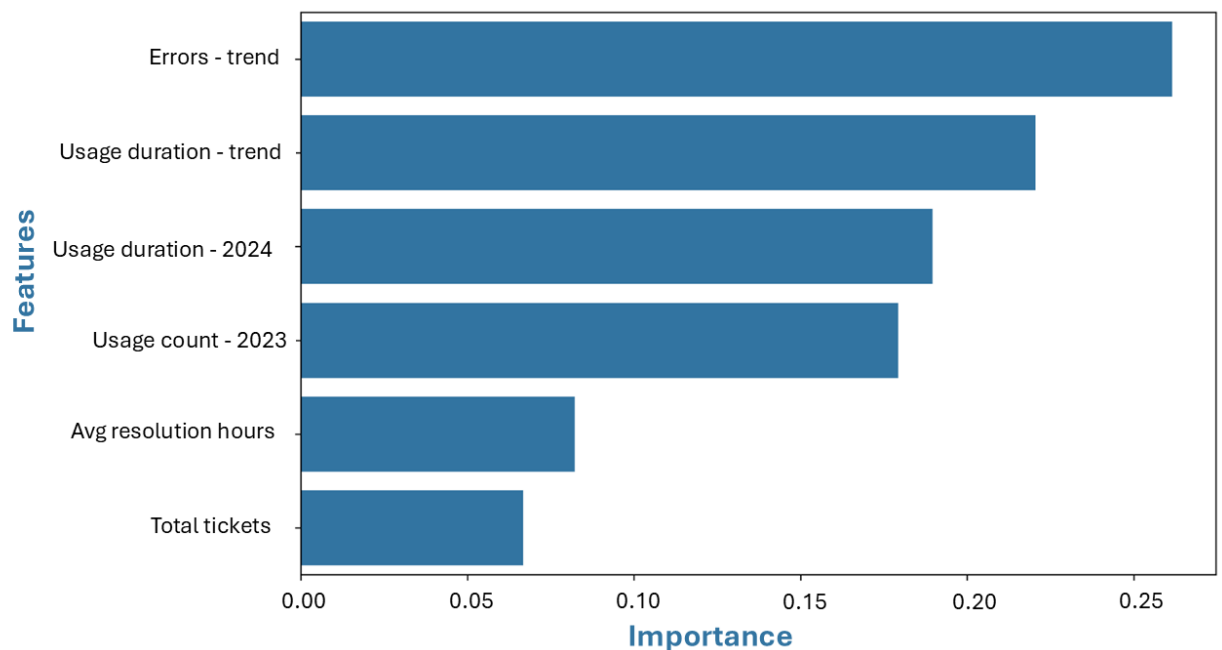
- KMeans clustering uses total ARR and seat count at 'subscription' level
 - Firmographic data did not reveal distinct customer groups
 - This is one of the most impactful problems caused by an artificial data set
- Two distinct clusters were revealed:
 - Cluster 0, labelled 'Premier', containing 78 accounts
 - Cluster 1, labelled 'Core', containing 422 accounts
- Subsequent predictive models leverage these segments for prioritisation and the assignment of resources for the resulting action plans

Health Scoring

- A model fully aligned with the Weiss framework was not possible due to data limitations. The framework suggests:
 - Product usage (depth and breadth): 30% ✓
 - Executive engagement frequency: 25% ✗
 - Support ticket volume and sentiment: 15% ✓
 - Contract renewal timeline: 10% ✗
 - User growth/adoption rates: 10% ✗
 - Invoice payment history: 10% ✗
- From the available data a first iteration of a health scoring framework was built as follows
 - Usage (50%)
 - Distinct features (Usage depth): 25%
 - Usage count (mean) (Usage breadth): 25%
 - Support (50%)
 - Satisfaction score (mean): 30%
 - Total tickets: 10%
 - Mean time to resolve: 10%
- Health scores range from 21 to 80 (average 66)
- The narrow range and absence of very low scores suggest synthetic data limitations, not a genuine distribution
- Premier average health marginally higher than Core accounts
- Core accounts show more low-scoring outliers (94). Prioritise these in the intervention plan

Churn Prediction

- Feature engineering did not find strong signals. This model is a v1 to refine as data improves
- The best results came from combining usage trends and cumulative support tickets



Churn model contributing features in order of importance

- Model's statistics:
 - **Baseline:** A 'lazy' model predicting no churn would achieve 77% accuracy but zero actual churn events
 - **Accuracy 57%** – this model predicts the right answer just over half the time
 - **Precision 26%** – accounts flagged have one in four chance of churning
 - **Recall 46%** – it catches nearly half of all customers who eventually churn
 - Performance restricted by artificial data, resulting in weak signals

- Revenue protected
 - High priority – Premier
 - Prioritise 38 high-risk accounts
 - This represents \$23.4M ARR
 - Captures \$6.1M at risk (based on 26% precision)
 - Medium priority (high probability core)
 - 17 customers > 70% probability
 - Represents \$920,000 ARR

Recommendations

Process review and internal alignment

- Assign named CSM resources to Premier accounts; pooled CSM resources to Core accounts
- Review and update playbooks to reflect data-driven intervention, for use in the phases listed below
- Notify Professional Services, Partner Network, and Customer Support to prepare for increased activity on flagged accounts

Just in time intervention plan

- **High priority (Premier segment, high-risk):**
 - 38 accounts representing \$23.4M ARR
 - CSM outreach within two weeks
- **Medium priority (Core segment, high-risk):**
 - 17 accounts representing \$920k ARR
 - Pooled CSM outreach within one month

- **Core low-scoring outliers (Health Scoring - 94 accounts):**
 - Added to weekly health review.
 - No immediate outreach unless risk escalates or health score drops further

Expansion plan

Consolidate Premier accounts onto a single Enterprise-wide contract. Analysis shows many hold multiple plan tiers, creating unnecessary support complexity and an inconsistent experience where users on different tiers compare access and features.

The value of consolidation is not primarily revenue uplift. It is:

- Simpler billing — one invoice, not multiple
- Unified feature access — no more 'why does she have that and I don't'
- Easier renewal management — one date, one negotiation
- Potential seat savings through consolidation

Further analysis would be required to quantify financial impact, including per-seat pricing and current usage data.

Limitations in the data

This analysis uses the RavenStack dataset (Rivalytics, 2025), built as a learning resource for data analysts. The artificial nature of the data was revealed in several places. Where it created blockers, I implemented workarounds. Where it exposed patterns likely to appear in the real world, I treated them as genuine problems and fixed them accordingly. The recommendations below assume real scenarios, not synthetic quirks.

- Firmographics
 - **Problem:** No distinct groupings were revealed by KMeans while exploring the firmographic data available for segmentation
 - **Impact:** It was necessary to build rudimentary segmentation limited to revenue and license volume
 - **Suggestion:** Review firmographics and align with industry best practice. Missing examples include: company size, ownership/status descriptions and growth/performance metrics
- Churn event data
 - **Problem:** No single churn driver emerged, with reason codes distributed across categories evenly. Additionally, the reason code conflicted with the feedback text in 88% of records
 - **Impact:** Historic data cannot be trusted for root cause analysis or to validate predictive models
 - **Suggestion:** Make reason codes mandatory, with an 'other' option which opens a free text field. Add regular cycles of analysis to check for consistency and update reason codes
- Support ticket data
 - **Problem:** Satisfaction scores ranged only from 3.0 to 5.0 (mean 3.97). No customer rated below 3. Ticket data was provided at account level, not subscription level, creating granularity mismatches.
 - **Impact:** The signal is compressed. Satisfied customers still churn, but the data cannot distinguish why. Account-level aggregation loses subscription-specific patterns.
 - **Suggestion:** I assumed the provided Support Ticket data was generated from a report by a busy Support Manager who misunderstood the request. Open Support Ticket data to analytics team to capture at Subscription level, ensuring realistic distributions of satisfaction scores and escalating should the documented issues repeat themselves in future analysis cycles

- Usage data
 - **Problem:** Duplicate 'usage_id' values linked to different 'subscription_id' values. Feature usage is sparse, with many features showing little or no adoption. Listed features appear hardly used.
 - **Impact:** Usage signals are weak. The model depends heavily on a small number of features. Sparse adoption limits the ability to identify engagement-driven churn.
 - **Suggestion:** Escalate and investigate problems encountered in usage data. Ensure v2 models contain unique keys and reliable feature usage data
- Executive engagement
 - **Problem:** The only satisfaction data available is captured from user feedback when support tickets are closed.
 - **Impact:** No indication of the intentions and sentiments of customers' decision makers
 - **Suggestion:** Survey fatigue is real. Ignored survey notifications deny vendors the ability to make predictions. Communications with customer stakeholders at all levels should be through a central tool so that conversations can be analysed with Natural Language Processing to gauge executive sentiment continuously
- Financial data
 - **Problem:** No meaningful financial/billing data was provided to be used in scoring frameworks and predictive models. Billing frequency and the 'auto_renew_flag' appear to be arbitrarily applied labels, giving weak signals
 - **Impact:** V1 churn model has relatively low accuracy, precision and recall. Health scoring tool has no invoice payment history in weighting framework
 - **Suggestion:** Include financial data in the data set to strengthen health scoring with renewal timing and invoice payment history and include in predictive models to pick up on the 'passive aggressive' signals of churn

Next investments

Analytical stack

RavenStack is evaluating solutions for company-wide analytical systems. In the meantime, a proof of concept (POC) will be built using existing low-cost infrastructure (PostgreSQL).

This analysis forms the foundation for dashboards, a feature store, and future model iterations. It currently exists only in notebooks.

Until an enterprise solution is ready, a data engineering "phase 2" will:

- Build a complete, end-to-end POC, from analysis to storage to scale
- Design an ETL pipeline using staged validation (medallion methodology) to ensure production reliability
- A 'star schema' will embed business logic into table structures, making the data easily accessible for:
 - AI analytics tools,
 - Dashboard platforms (Tableau, Power BI etc)
 - Other analysts continuing this work in my absence